Private Capital and Sport

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A note on "post-money equity valuation"

A **post-money equity valuation** is the value of the company's equity *immediately after* new capital has been invested.

In the case used below, the fund invests $150 million for a 15% ownership stake. If this is a **primary investment**—meaning the $150 million goes into the club in exchange for newly issued equity—then the implied post-money equity valuation is

$$\frac{150}{0.15} = \$1{,}000 \text{ million}$$

The corresponding pre-money equity valuation is

$$\$1{,}000 \text{ million} - \$150 \text{ million} = \$850 \text{ million}$$

If, instead, the fund is buying existing shares from a current owner and no new money enters the club, "post-money" is not the most precise term. In that case, it is better to refer to the **implied equity valuation** of the club.

For the problems below, we assume the investment is primary and therefore use the $1.0 billion post-money equity valuation given in the case.

Problem 1 — Does the expected exit justify the entry price?

We want to determine whether the proposed minority investment generates an annualized return that meets the private-capital fund's required return.

The case assumptions are:

- The fund invests $150 million today.
- In exchange, it receives 15% of the club's equity.
- The implied post-money equity valuation of the club is $1.0 billion.
- The planned holding period is six years.
- The expected club equity valuation at the end of year 6 is $1.7 billion.
- For the first version of the problem, assume no interim cash distributions and no dilution of the 15% stake.

- For the classroom exercise, assume the fund requires a **15% annualized return**. This 15% is an illustrative underwriting assumption used to create a decision threshold; it is not presented as a universal private-equity hurdle rate.

What is the stake of the investor expected to be worth at exit?

Let

- $\alpha = 0.15$, be the investor's ownership fraction
- $v_0 = \$1{,}000$ million, be the club's post-money equity valuation at the time of investment
- i_0 be the cash invested by the fund at time 0
- $v_6 = \$1{,}700$ million, be the expected equity valuation of the entire club at the end of year 6
- x_6 be the cash proceeds the investor would receive from selling its 15% stake at the end of year 6

Assuming no dilution and a sale at full pro-rata value,

$$x_6 = \alpha v_6$$

Therefore,

$$x_6 = 0.15 \times 1{,}700 = \$255 \text{ million}$$

So the simplified cash-flow pattern is

$$-\$150 \text{ million at } t = 0 \qquad \longrightarrow \qquad +\$255 \text{ million at } t = 6$$

The stake of the investor at exit comes from translating the expected club-level exit valuation into the cash proceeds attributable to the minority stake of the investor.

What is the investment multiple?

The multiple of invested capital (MOIC) is

$$\text{moic} = \frac{\text{total value received}}{\text{capital invested}}$$

For this deal,

$$\text{moic} = \frac{x_6}{i_0} = \frac{255}{150} = 1.70\times$$

Every \$1 invested is expected to become \$1.70 at exit. MOIC is a static measure with no time dimension. It captures the multiple of invested capital generated, but not the speed at which that value is created. A 1.70× outcome over three years is economically different from a 1.70× outcome over six years.

What annualized return is implied by the proposed deal?

Let

- r_{deal} be the annualized return implied by the proposed investment and expected exit proceeds

Because there is one cash outflow today and one cash inflow six years from now, the annualized return satisfies

$$i_0(1 + r_{\text{deal}})^6 = x_6$$

Substituting the values,

$$150(1 + r_{\text{deal}})^6 = 255 \longrightarrow (1 + r_{\text{deal}})^6 = \frac{255}{150} = 1.70 \longrightarrow r_{\text{deal}} = 1.70^{1/6} - 1 \approx 0.0925$$

Hence,

$$r_{\text{deal}} \approx 9.25\% \text{ per year}$$

Therefore, the annualized return implied by the proposed deal is obtained by converting the six-year investment multiple into an annualized compound return. Because this simplified case has only one initial outflow and one terminal inflow, this annualized compound return is also the deal's IRR.

Does the expected deal return meet the fund's required return?

Let

- $k = 15\%$ be the the annualized return the investor requires before accepting the deal, while
- r_{deal} is the annualized return generated by the deal assumptions

We have

$$r_{\text{deal}} = 9.25\% \qquad \text{and} \qquad k = 15\%$$

Therefore,

$$r_{\text{deal}} < k$$

Under the current assumptions—\$150 million invested today, no interim distributions, and \$255 million of exit proceeds in year 6—the proposed investment does not meet the fund's assumed 15% annual required return.

What exit proceeds would be required to earn 15% per year?

Let

- x_6^{req} be the year-6 proceeds required for a \$150 million investment to earn the required annualized return $k = 15\%$

The required terminal proceeds satisfy

$$x_6^{\text{req}} = i_0(1+k)^6$$

Thus,

$$x_6^{\text{req}} = 150(1.15)^6 \approx \$346.97 \text{ million}$$

Because the investor owns 15%, the required club equity valuation at exit is

$$v_6^{\text{req}} = \frac{x_6^{\text{req}}}{\alpha} = \frac{346.97}{0.15} \approx \$2.31 \text{ billion}$$

If there are no interim distributions, the club would need to be worth approximately \$2.31 billion in year 6—not \$1.7 billion—for the investor to earn 15% per year on a \$150 million investment.

Equivalent underwriting question: what is the maximum amount the investor should pay today?

Suppose

- the expected year-6 proceeds remain $x_6 = \$255$ million
- and the required annualized return remains $k = 15\%$
- $i_{0,\max}$ be the maximum amount the investor can pay today while still earning 15% per year

Discount the expected exit proceeds back six years:

$$i_{0,\max} = \frac{x_6}{(1+k)^6}$$

Therefore,

$$i_{0,\max} = \frac{255}{(1.15)^6} \approx \$110.24 \text{ million}$$

The corresponding maximum post-money club valuation is

$$v_{0,\max} = \frac{i_{0,\max}}{\alpha} = \frac{110.24}{0.15} \approx \$735 \text{ million}$$

The seller's proposed \$1.0 billion post-money valuation is above the approximately \$735 million valuation consistent with a 15% annual required return under the current exit assumptions.

What could close the return gap?

The simplified problem has so far assumed no cash distributions before exit. Let

- d_t be the cash distribution received by the investor in year t

If distributions occur during the holding period, the present value of the investor's expected cash flows is

$$v_{\text{investment}} = \sum_{t=1}^{6} \frac{d_t}{(1+k)^t} + \frac{x_6}{(1+k)^6}$$

The net present value is

$$npv = v_{\text{investment}} - i_0$$

The investment satisfies the required-return threshold when

$$npv \geq 0$$

Thus, a higher entry valuation can be justified only if the investment thesis contains additional value—for example, interim distributions, a higher exit valuation, a shorter holding period, or another credible source of cash flow.

Problem 2 — What is the economic cost of restricted exit liquidity?

Problem 1 assumed that the fund could sell its 15% stake in year 6 for its full pro-rata value of $255 million. That assumption is particularly strong in professional sport. A minority investor may face a limited buyer pool, league approval requirements, transfer restrictions, weak control rights, or a minority discount at exit. Therefore, this problem answers the question:

> How does restricted exit liquidity change the present value of the same 15% stake?

The purpose of the model is to show that ownership percentage alone does not determine investment value. Exit rights and transfer constraints affect the amount of cash the investor can realistically monetize.

From Problem 1,

- the expected equity valuation of the club in year 6 is $v_6 = \$1{,}700$ million
- the investor owns $\alpha = 0.15$
- the full pro-rata value of the investor's stake is $x_6 = \alpha v_6 = \$255$ million

This is the amount the investor would receive if it can sell at full proportional value.

Introduce the probability of a full-value exit

Let

- the probability that the investor can execute an approved sale at the end of year 6 be $q = 0.70$
- the 70% probability be an illustrative assumption, not an empirical estimate of league approval probabilities

with probability $q = 0.70$, the investor receives the full pro-rata value of the stake, $x_6 = \$255$ million

Define the restricted-exit outcome

If a full-value exit cannot be executed, assume the investor can still monetize the stake in year 6, but only at a discount to its pro-rata value.

Let

- the fractional haircut applied in the restricted-exit state be $h = 0.25$. A 25% haircut means that the investor receives 75% of the full pro-rata value
- the year-6 cash proceeds in the restricted-exit state be $x_6^R = (1 - h)x_6$

therefore,

$$x_6^R = (1 - 0.25)(255) = \$191.25 \text{ million}$$

These formulas translate the limited liquidity into a cash-flow consequence rather than treating "illiquidity" as an abstract label.

Calculate expected year-6 exit proceeds

There are now two possible year-6 outcomes:

- With probability q, the investor exits at full pro-rata value x_6.
- With probability $1 - q$, the investor exits at the restricted value x_6^R.

Therefore expected year-6 proceeds are

$$\mathbb{E}[x_6] = qx_6 + (1 - q)x_6^R.$$

Because

$$x_6^R = (1 - h)x_6$$

we can write

$$\mathbb{E}[x_6] = qx_6 + (1 - q)(1 - h)x_6 \quad \longrightarrow \quad \mathbb{E}[x_6] = x_6 \left[1 - h(1 - q)\right]$$

Substituting the values,

$$\mathbb{E}[x_6] = 255 \left[1 - 0.25(1 - 0.70)\right] = \$235.875 \text{ million}$$

Exit restrictions reduce the expected cash proceeds from \$255 million to approximately \$235.88 million even though the club's expected year-6 equity valuation has not changed.

Put exit risk into the present-value calculation

Using $k = 15\%$ as the fund's required annualized return, if there are still no interim distributions, the present value of expected exit proceeds is

$$pv_{\text{exit}} = \frac{\mathbb{E}[x_6]}{(1 + k)^6}$$

Thus,

$$pv_{\text{exit}} = \frac{235.875}{(1.15)^6} \approx \$101.98 \text{ million}$$

The corresponding NPV is

$$npv = pv_{\text{exit}} - i_0 == 101.98 - 150 \approx -\$48.02 \text{ million}$$

Using the cash the investor can realistically expect to monetize changes the investment value.

Measure the value lost because of exit restrictions

Under the unrestricted assumption from Problem 1, the present value of the \$255 million year-6 proceeds is

$$pv_{\text{unrestricted}} = \frac{255}{(1.15)^6} \approx \$110.24 \text{ million}$$

Under the restricted-exit assumptions,

$$pv_{\text{restricted}} \approx \$101.98 \text{ million}$$

Therefore the present-value loss created by the exit restriction is

$$\Delta pv = pv_{\text{restricted}} - pv_{\text{unrestricted}} \approx -\$8.27 \text{ million}$$

The same result can be written directly as

$$\text{exit-risk cost} = \frac{x_6 h(1-q)}{(1+k)^6}$$

The economic cost of illiquidity is the present value of the cash the investor expects to lose because the stake may not be monetized at full pro-rata value.

Comparative statics: what makes the minority stake more valuable?

The present value of expected exit proceeds is

$$pv_{\text{exit}} = \frac{x_6[1-h(1-q)]}{(1+k)^6}$$

differentiating with respect to q:

$$\frac{\partial pv_{\text{exit}}}{\partial q} = \frac{x_6 h}{(1+k)^6} > 0$$

therefore, as we would logically expect, holding everything else constant, a higher probability of achieving a full-value exit increases the value of the stake.

Differentiating with respect to h:

$$\frac{\partial pv_{\text{exit}}}{\partial h} = -\frac{x_6(1-q)}{(1+k)^6} < 0$$

therefore, holding everything else constant, a larger restricted-exit discount reduces the value of the stake.

Transfer rights, tag-along rights, information rights, approved-buyer access, and other governance protections matter because they can affect the ability of the investor to monetize the stake and the price obtained at exit. The legal terms therefore have an economic value.

Recap

Problem 1 asks whether the **entry price, holding period, and expected exit valuation** generate the fund's required annualized return.

Problem 2 then relaxes the unrealistic assumption that the investor can automatically convert its pro-rata stake value into cash at exit. It shows mathematically why **liquidity, transfer restrictions, league approval, and governance rights** must enter the underwriting analysis.

The progression is therefore

$$\text{entry price} \longrightarrow \text{exit value} \longrightarrow \text{MOIC and annualized return} \longrightarrow \text{required return}$$
$$\longrightarrow \text{exit restrictions} \longrightarrow \text{economic value of governance rights.}$$

Source note

The numerical deal setup follows the Week 3 case on a 15% minority investment in a professional club. The 15% required annualized return, the 70% full-value exit probability, and the 25% restricted-exit haircut are explicitly introduced here as classroom assumptions for the mathematical exercise; they are not empirical estimates from the source materials.